



TECHBIZACADEMY · ETHICAL HACKING INTERNSHIP
Week 2 · Task 4

Linux Fundamentals

Commands · Files & Directories · Permissions

Name	Mah Noor
Student ID	TBS-2026-242
Cohort	EH-2026-02
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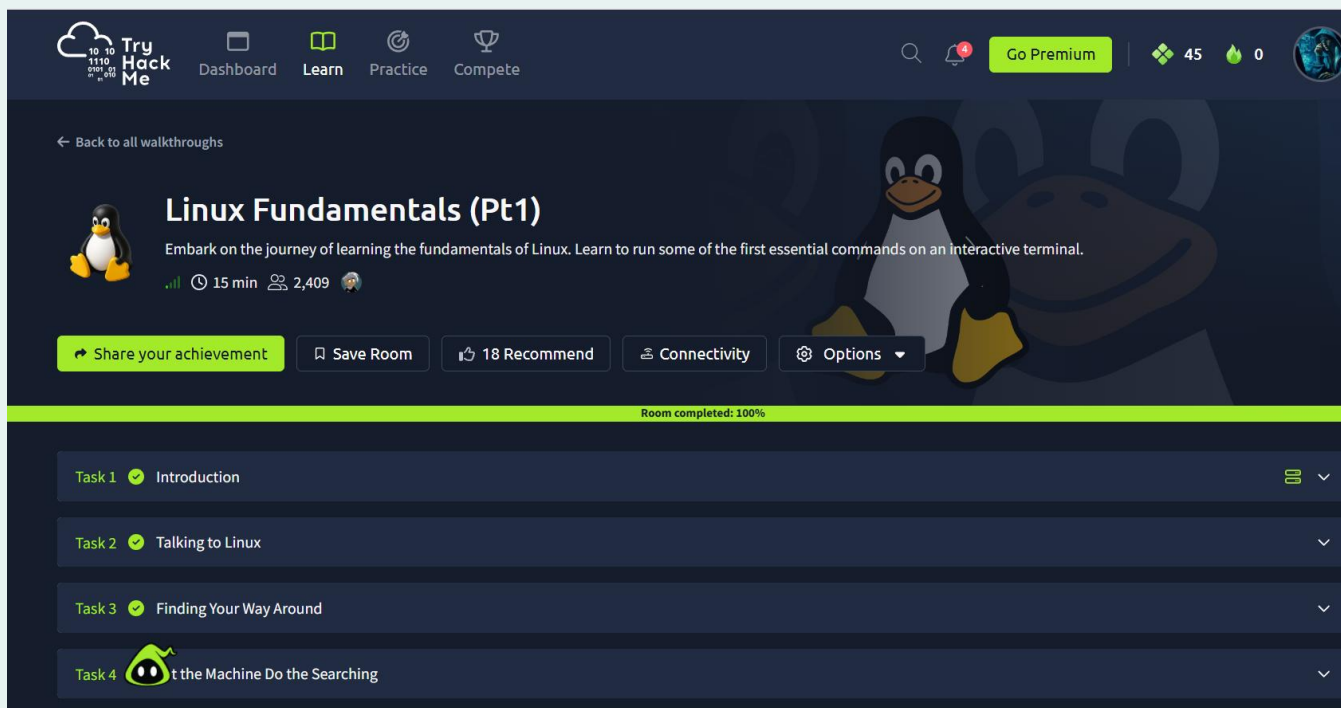
Kali Linux · File System & Permissions · Package Management

1. Introduction

This report documents the Day 1 practical work and Day 2 write-up for the Linux Fundamentals task. It covers navigating the Linux file system, running basic commands, creating and reading files, managing permissions with `chmod/chown`, and installing a tool using `apt` on Kali Linux.

2. TryHackMe Room Completion Proof

Room: Linux Fundamentals Part 1 (free tier, mandatory).



Description: TryHackMe room completion screen showing 100% progress / completed status for 'Linux Fundamentals Part 1', with your username visible.

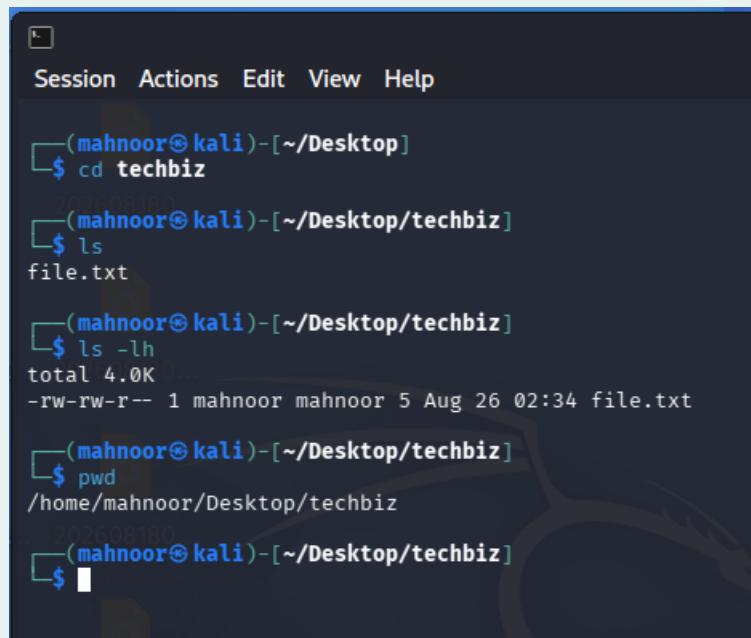
3. Command Cheat-Sheet & Practical Screenshots

Each command below is listed with its purpose, followed by a placeholder for the corresponding terminal screenshot.

Screenshot 1 · Navigation Basics — `pwd`, `ls`, `ls -l`, `cd`

Command(s): `pwd` | `ls` | `ls -l` | `cd <directory>`

Purpose: `pwd` shows the current working directory; `ls` lists directory contents; `ls -l` shows a long listing with permissions, owner, and size; `cd` changes the current directory.

A terminal window with a dark background and light blue text. The window has a menu bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'. The prompt is '(mahnoor@kali)-[~/Desktop]'. The user enters '\$ cd techbiz'. The prompt changes to '(mahnoor@kali)-[~/Desktop/techbiz]'. The user enters '\$ ls', and the output is 'file.txt'. The user enters '\$ ls -lh', and the output is 'total 4.0K' followed by a line ' -rw-rw-r-- 1 mahnoor mahnoor 5 Aug 26 02:34 file.txt'. The user enters '\$ pwd', and the output is '/home/mahnoor/Desktop/techbiz'. The user enters '\$' and the prompt returns to '(mahnoor@kali)-[~/Desktop/techbiz]'.

```
(mahnoor@kali)-[~/Desktop]
$ cd techbiz

(mahnoor@kali)-[~/Desktop/techbiz]
$ ls
file.txt

(mahnoor@kali)-[~/Desktop/techbiz]
$ ls -lh
total 4.0K
-rw-rw-r-- 1 mahnoor mahnoor 5 Aug 26 02:34 file.txt

(mahnoor@kali)-[~/Desktop/techbiz]
$ pwd
/home/mahnoor/Desktop/techbiz

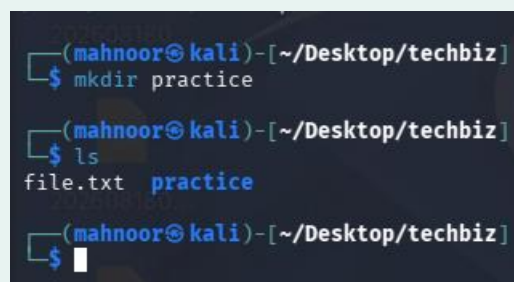
(mahnoor@kali)-[~/Desktop/techbiz]
$
```

Description: Terminal showing pwd output, a directory listing (ls and ls -l), and moving into another directory with cd.

Screenshot 2 • Create a Folder — mkdir

Command(s): mkdir practice_folder

Purpose: Creates a new directory to organize the day's practice files.

A terminal window with a dark background and light blue text. The prompt is '(mahnoor@kali)-[~/Desktop/techbiz]'. The user enters '\$ mkdir practice'. The prompt changes to '(mahnoor@kali)-[~/Desktop/techbiz]'. The user enters '\$ ls', and the output is 'file.txt practice'. The user enters '\$' and the prompt returns to '(mahnoor@kali)-[~/Desktop/techbiz]'.

```
(mahnoor@kali)-[~/Desktop/techbiz]
$ mkdir practice

(mahnoor@kali)-[~/Desktop/techbiz]
$ ls
file.txt practice

(mahnoor@kali)-[~/Desktop/techbiz]
$
```

Description: Terminal showing the mkdir command and ls confirming the new folder exists.

Screenshot 3 • Create & Write to a File

Command(s): cd practice_folder && nano notes.txt (or: echo "Hello Linux" > notes.txt)

Purpose: Creates a new text file inside the folder and writes some text into it.

```
(mahnoor@kali) - [~/Desktop/techbiz]
$ cd practice

(mahnoor@kali) - [~/Desktop/techbiz/practice]
$ nano notes.txt

(mahnoor@kali) - [~/Desktop/techbiz/practice]
$ echo "Hello Linux" > notes.txt
```

Description: Terminal/editor showing the file being created and text typed/saved inside it.

Screenshot 4 · Read a File — cat

Command(s): cat notes.txt

Purpose: Displays the contents of the file directly in the terminal to confirm the text was saved correctly.

```
(mahnoor@kali) - [~/Desktop/techbiz/practice]
$ cat notes.txt
Hello Linux
```

Description: Terminal showing the cat command and the file's text content printed below it.

Screenshot 5 · Copy & Move Files — cp, mv

Command(s): cp notes.txt notes_backup.txt | mv notes_backup.txt ../notes_backup.txt

Purpose: cp duplicates a file; mv moves (or renames) a file to a new location.

```
(mahnoor@kali) - [~/Desktop/techbiz/practice]
$ cp notes.txt notes_backup.txt | mv notes_backup.txt ../notes_backup.txt

(mahnoor@kali) - [~/Desktop/techbiz/practice]
$ ls
notes.txt

(mahnoor@kali) - [~/Desktop/techbiz/practice]
$ cd ..

(mahnoor@kali) - [~/Desktop/techbiz]
$ ls
file.txt  notes_backup.txt  practice
```

Description: Terminal showing cp creating a copy, mv relocating/renaming it, and ls confirming the change.

Screenshot 6 • Delete a File — rm

Command(s): `rm notes_backup.txt`

Purpose: Removes a file that is no longer needed.

```
(mahnoor@kali)-[~/Desktop/techbiz]
$ rm notes_backup.txt

(mahnoor@kali)-[~/Desktop/techbiz]
$ ls
file.txt  practice
```

Description: Terminal showing the rm command and ls confirming the file is gone.

Screenshot 7 • Change Permissions — chmod

Command(s): `ls -l notes.txt → chmod 755 notes.txt → ls -l notes.txt`

Purpose: chmod changes read/write/execute permissions on a file. 755 gives the owner full access and read+execute to group/others.

```
(mahnoor@kali)-[~/Desktop/techbiz]
$ cd practice

(mahnoor@kali)-[~/Desktop/techbiz/practice]
$ ls -l notes.txt
-rw-rw-r-- 1 mahnoor mahnoor 12 Aug 26 02:42 notes.txt

(mahnoor@kali)-[~/Desktop/techbiz/practice]
$ chmod 755 notes.txt

(mahnoor@kali)-[~/Desktop/techbiz/practice]
$ ls -l notes.txt
-rwxr-xr-x 1 mahnoor mahnoor 12 Aug 26 02:42 notes.txt

(mahnoor@kali)-[~/Desktop/techbiz/practice]
$
```

Description: Two ls -l outputs side by side (or before/after) showing the permission bits changing after chmod 755.

Screenshot 8 • Change Ownership — chown

Command(s): `sudo chown $USER:$USER notes.txt`

Purpose: Changes the user/group owner of a file — useful when a file needs to be reassigned to a different user or group.

```
(mahnoor@kali)~[~/Desktop/techbiz/practice]
$ sudo chown $USER:$USER notes.txt
[sudo] password for mahnoor:

(mahnoor@kali)~[~/Desktop/techbiz/practice]
$ ls -l notes.txt
-rwxr-xr-x 1 mahnoor mahnoor 12 Aug 26 02:42 notes.txt
```

Description: Terminal showing the chown command running successfully (sudo password prompt can be blurred) and ls -l confirming the new owner.

Screenshot 9 • Install a Package — sudo apt install

Command(s): `sudo apt update && sudo apt install nmap`

Purpose: apt is Kali's package manager; this installs the nmap network scanning tool from the repositories.

```
(mahnoor@kali)~[~/Desktop/techbiz/practice]
$ sudo apt update && sudo apt install nmap
[sudo] password for mahnoor:
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.4 MB]
```

Description: Terminal showing the apt install process completing successfully (or 'nmap is already the newest version').

Screenshot 10 • Verify Installation

Command(s): `nmap --version`

Purpose: Confirms that nmap was installed correctly and is ready to use.

```
(mahnoor@kali)~[~/Desktop/techbiz/practice]
$ nmap --version
Nmap version 7.99 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
```

Description: Terminal showing the nmap version output.

4. Additional Commands Practiced (20 total)

List each command you practiced with a one-line note on its output/purpose. Example format below — replace with your own.

- ❖ `whoami` — shows the current logged-in user
- ❖ `uname -a` — shows kernel and system information
- ❖ `df -h` — shows disk space usage in human-readable form
- ❖ `top / htop` — shows running processes and resource usage
- ❖ `ps aux` — lists all running processes
- ❖ `history` — shows previously run commands
- ❖ `man <command>` — shows the manual page for a command
- ❖ `clear` — clears the terminal screen
- ❖ `touch <file>` — creates an empty file
- ❖ `grep <pattern> <file>` — searches for text inside a file
- ❖ `find / -name <file>` — searches for a file across the filesystem
- ❖ `echo <text>` — prints text to the terminal
- ❖ `ifconfig / ip a` — shows network interface information
- ❖ `ping <host>` — tests network connectivity to a host
- ❖ `whatis <command>` — shows a one-line description of a command
- ❖ `which <command>` — shows the path of a command's executable
- ❖ `date` — shows the current system date and time
- ❖ `hostname` — shows the machine's hostname
- ❖ `cat /etc/os-release` — shows OS/distribution details
- ❖ `sudo -l` — lists sudo privileges for the current user

5. Directory Reference Notes

`/` — root of the filesystem, top of the directory tree.

`/home` — contains personal directories for each user.

`/etc` — system-wide configuration files.

`/var` — variable data such as logs, mail, and spool files.

6. Conclusion

Through this task I practiced core Linux navigation and file management commands, created and read files, adjusted permissions with `chmod` and `chown`, and installed a tool using `apt` on Kali Linux — building a working foundation for the ethical hacking commands used in later tasks.